TechPost ELEVATED

DC – AI – Routing – Switching – Security – Wireless – Automation

Telemetry Collector and Graphical Frontend on Junos Evolved

An HPE Juniper Networking Techpost

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Contents

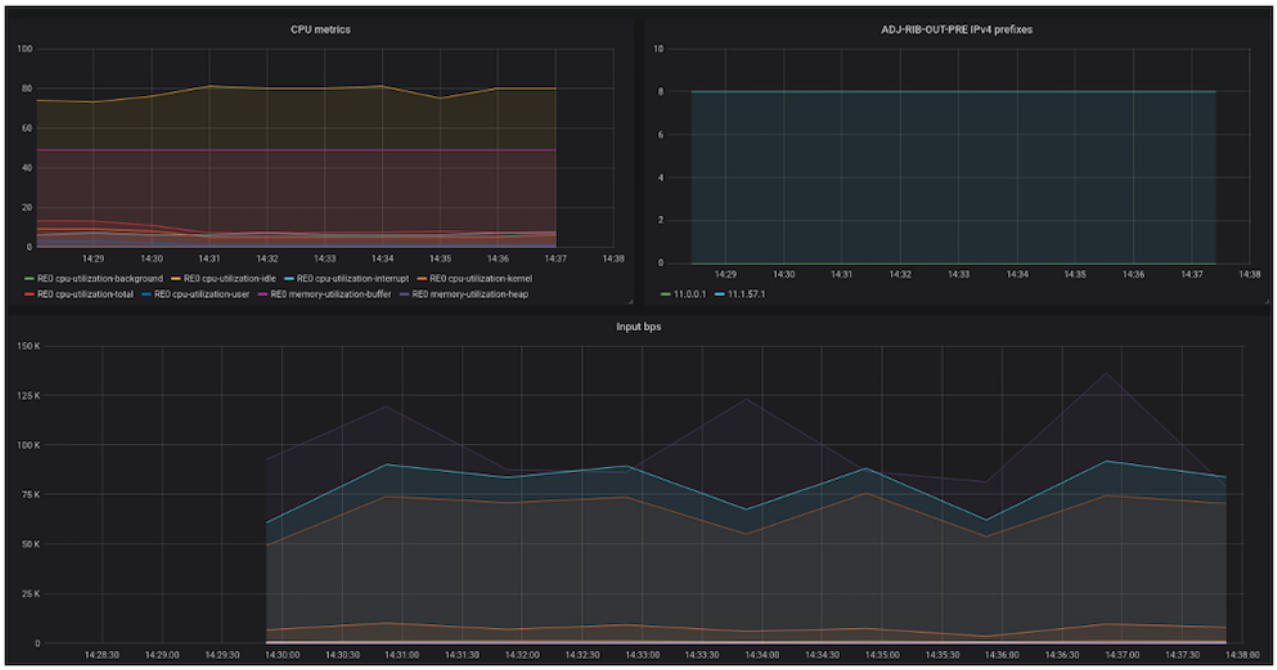
1 Telemetry Collector and Graphical Front End on Junos Evolved 2

1.1 Introduction . . . 2
1.2 Prepare Junos Evolved for Streaming Telemetry . . . 2
1.3 Create Configuration Files . . . 3
1.4 Start Docker Containers . . . 4
1.5 Work with a Graphical Interface . . . 4
1.6 Resources Management . . . 5
1.7 Useful links . . . 6
1.8 Glossary . . . 6

1 Telemetry Collector and Graphical Front End on Junos Evolved

Anton Elita - 07/18/2022





1.1 Introduction

- Configure Junos to accept Remote Procedure Calls (gRPC)
- Prepare configuration files for docker containers
- Start Docker containers
- Access the graphical interface and create own dashboards for interesting telemetry streams

1.2 Prepare Junos Evolved for Streaming Telemetry

Junos Evolved 21.4R1 is used here as an example. Streaming measurements out of a networking node can be performed in a few ways. In this guide, OpenConfig gRPC Network Management Interface (GNMI) is used for data encoding and transport. It is a widely adopted choice.

Basic configuration of gRPC:

```
1  system {
2     management-instance;
3     services {
```

```
        extension-service {
            request-response {
                grpc {
                    clear-text {
                        port 32767;
                    }
                    max-connections 30;
                    routing-instance mgmt_junos;
                }
            }
        }
    }
}
routing-instances {
    mgmt_junos {
        description default;
    }
}
```

```
> start shell user root
# export DOCKER_HOST=unix:///run/docker-mgmt_junos.sock
# docker image pull grafana/grafana:5.4.5
# docker image pull influxdb:1.8.10
# docker image pull telegrafa
```

```
docker image pull grafana/grafana:5.4.5
docker image pull influxdb:1.8.10
docker image pull telegraf
docker save grafana/grafana -o /var/tmp/grafana.tar
docker save influxdb:1.8.10 -o /var/tmp/influxdb.tar
docker save telegraf -o /var/tmp/telegraf.tar
scp /var/tmp/grafana.tar /var/tmp/influxdb.tar /var/tmp/telegraf.tar 11.254.253.7:/var/tmp/
```

```
> start shell user root
# export DOCKER_HOST=unix:///run/docker-mgmt_junos.sock
# docker load < /var/tmp/telegraf.tar
# docker load < /var/tmp/grafana.tar
# docker load < /var/tmp/influxdb.tar
```

```
> start shell user root
# mkdir -p /var/home/root/grafana/dashboards /var/home/root/telegraf /var/home/root/influxdb
# chmod -R 777 /var/home/root/grafana/
```

1.3 Create Configuration Files

```
> start shell user root
# cat <<EOF > /var/home/root/influxdb/env
INFLUXDB_DB=telegraf
INFLUXDB_USER=telegraf
INFLUXDB_ADMIN_ENABLED=true
INFLUXDB_ADMIN_USER=admin
INFLUXDB_ADMIN_PASSWORD=lab123
EOF
```

```
> start shell user root
# cat <<EOF > /var/home/root/telegraf/conf
[[outputs.influxdb]]
  urls = ["http://10.100.3.120:8086"]
  database = "telegraf"
  write_consistency = "any"
  timeout = "5s"
  username = "admin"
  password = "lab123"
[[inputs.jti_openconfig_telemetry]]
 servers = ["10.100.3.120:32767"]
 username = "user"
 password = "SECRET-PASSWORD"
 client_id = "telegraf"
 sample_frequency = "60000ms"
 sensors = [
    "60000ms bgp /network-instances/network-instance/protocols/protocol/bgp/neighbors/neighbor/
        state/session-state",
   "60000ms task /junos/task-memory-information/",
```

```
    "60000ms num_routes /bgp-rib/afi-safis/afi-safi/ipv4-unicast/neighbors/neighbor/adj-rib-out-pre/num-routes",
    "60000ms re0 /components/component[contains(name,'Engine0')]/properties/property[contains(name,'utilization')]/state/value/",
    "60000ms interfaces /interfaces/interface/"
  ]
str_as_tags = false
[[processors.converter]]
  [processors.converter.fields]
    float = ["/components/component/properties/property/state/value"]
EOF
```

1.4 Start Docker Containers

```
> start shell user root
# export DOCKER_HOST=unix:///run/docker-mgmt_junos.sock

# docker run \
-d --name grafana \
-v /var/home/root/grafana/:/var/lib/grafana/ \
-v /var/home/root/grafana/dashboards:/usr/share/grafana/conf/provisioning/dashboards/ \
-v /etc/localtime:/etc/localtime:ro \
--cap-add=NET_ADMIN \
--network=host \
--restart=always \
grafana/grafana:5.4.5

# docker run \
-d --name influxdb \
--env-file=/var/home/root/influxdb/env \
-v /etc/localtime:/etc/localtime:ro \
-v /var/home/root/influxdb:/var/lib/influxdb \
--cap-add=NET_ADMIN \
--network=host \
--restart=always \
influxdb:1.8.10

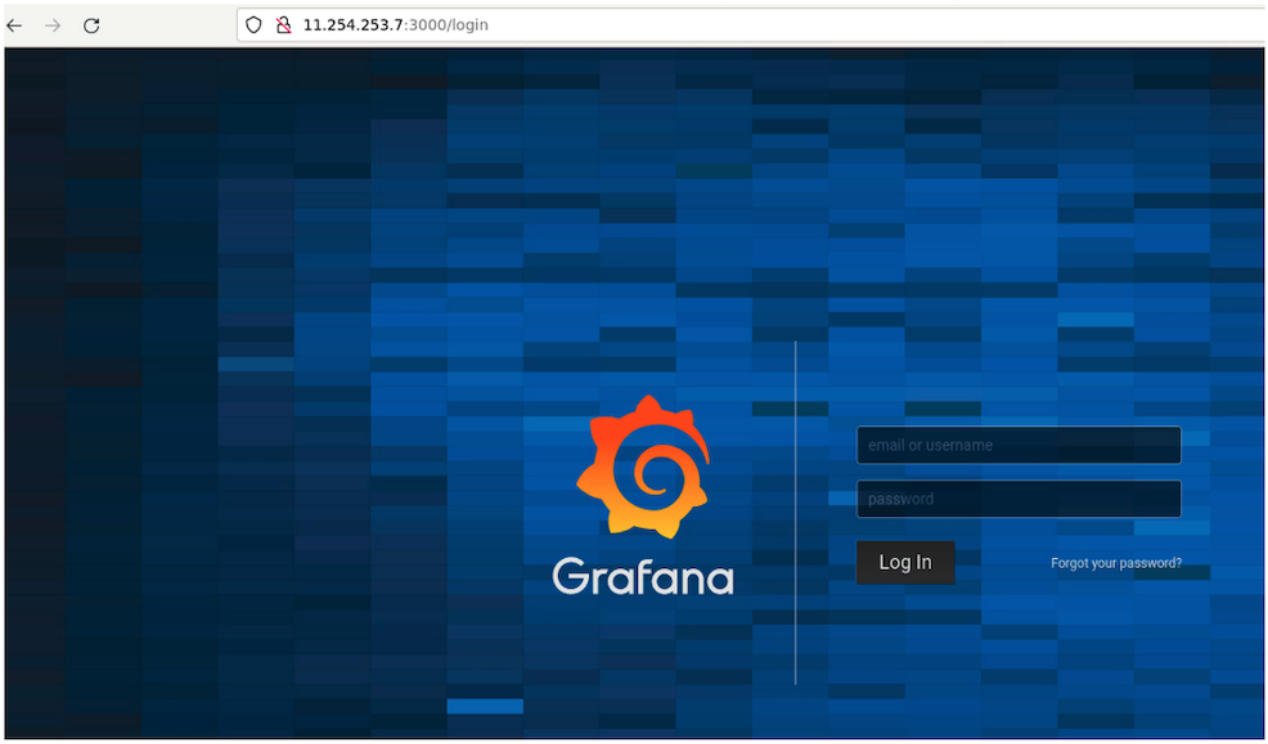
# docker run \
-d --name telegraf \
-v /var/home/root/telegraf/conf:/etc/telegraf/telegraf.conf:ro \
-v /etc/localtime:/etc/localtime:ro \
--cap-add=NET_ADMIN \
--network=host \
--restart=always \
telegraf
```

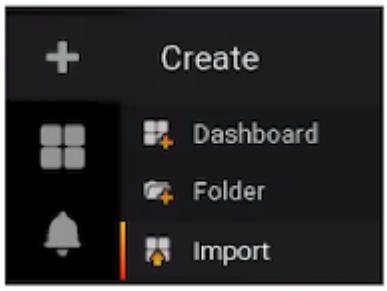
```
# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
8c105cb2a587 telegraf "/entrypoint.sh tele..." 21 hours ago Up 21 hours telegraf
332e790beb6c influxdb:1.8.10 "/entrypoint.sh infl..." 9 days ago Up 9 days influxdb
a40d11a0c2f0 grafana/grafana:5.4.5 "/run.sh" 9 days ago Up 9 days Grafana
```

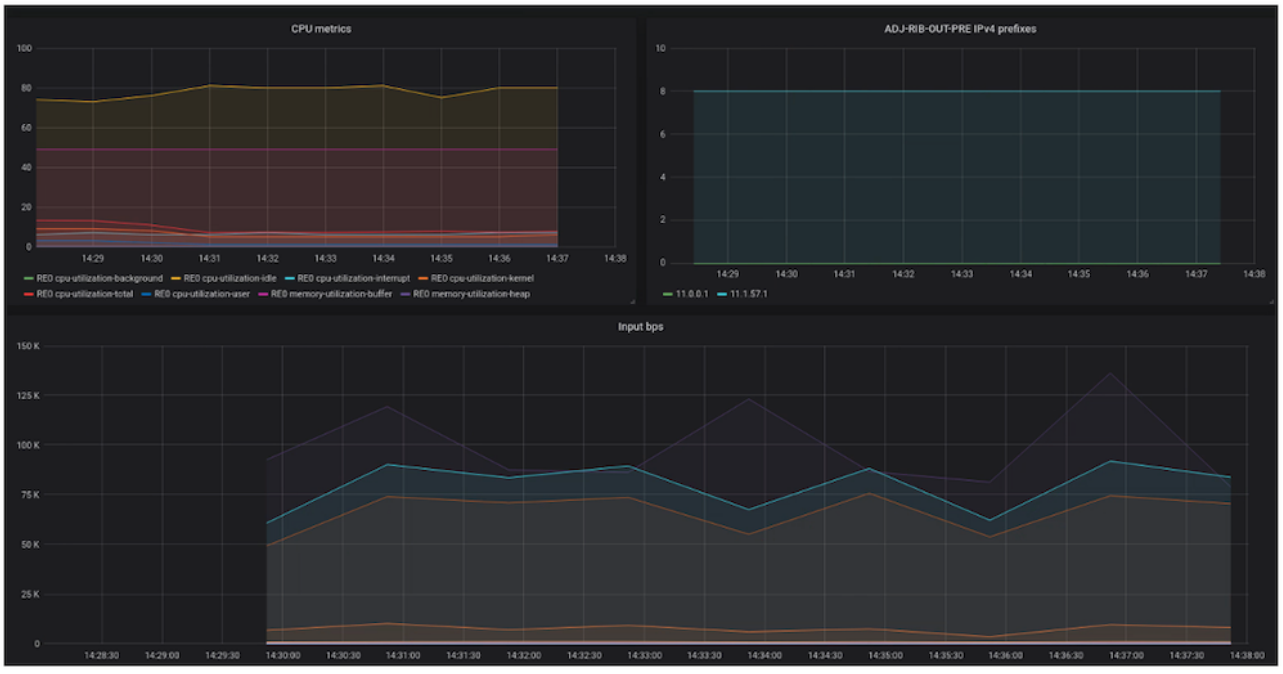
```
# docker stats --no-stream
CONTAINER ID NAME CPU % MEM USAGE / LIMIT MEM % NET I/O BLOCK I/O PIDS
8c105cb2a587 telegraf 1.20% 30.96MiB / 15.39GiB 0.20% 0B / 0B 0B / 0B 0
332e790beb6c influxdb 0.15% 265.3MiB / 15.39GiB 1.68% 0B / 0B 0B / 0B 0
a40d11a0c2f0 grafana 0.02% 20.56MiB / 15.39GiB 0.13% 0B / 0B 0B / 0B 0
```

1.5 Work with a Graphical Interface

```
http://11.254.253.7:3000/
```







1.6 Resources Management

```
1  > start shell user root
```

```
# export DOCKER_HOST=unix:///run/docker-mgmt_junos.sock

# cat /etc/extensions/platform_attributes
## Edit to change upper cap of total resource limits for all containers.
## applies only to containers and does not apply to container runtimes.
## memory.memsw.limit_in_bytes = EXTENSIONS_MEMORY_MAX_MIB + EXTENSIONS_MEMORY_SWAP_MAX_MIB:-0
## check current defaults, after starting extensions-cglimits.service
## $ /usr/libexec/extensions/extensions-cglimits get
## please start extensions-cglimits.service to apply changes here

## device size limit will be ignored once extensionsfs device is created
#EXTENSIONS_FS_DEVICE_SIZE_MIB=
#EXTENSIONS_CPU_QUOTA_PERCENTAGE=
#EXTENSIONS_MEMORY_MAX_MIB=
#EXTENSIONS_MEMORY_SWAP_MAX_MIB=
```

- **EXTENSIONS_FS_DEVICE_SIZE_MIB=** is the maximum storage space in bytes that containers can use. The default value is 8 GB or 30% of the total size of /var, whichever is smaller.
- **EXTENSIONS_CPU_QUOTA_PERCENTAGE=** is the maximum percentage of CPU usage that containers can use. The default value is 20% max CPU use across all cores.
- **EXTENSIONS_MEMORY_MAX_MIB=** is the maximum amount of physical memory in bytes that containers can use. The default value is 2 GB or 10% of total physical memory, whichever is smaller.

```
> start shell user root
# export DOCKER_HOST=unix:///run/docker-mgmt_junos.sock

# systemctl restart extensions-cglimits.service
```

```
> start shell user root
# export DOCKER_HOST=unix:///run/docker-mgmt_junos.sock

# docker exec -it influxdb influx -database telegraf -execute "show retention policies"
name duration shardGroupDuration replicaN default
---- -------- ------------------ -------- -------
autogen 0s 168h0m0s 1 true

# docker exec -it influxdb influx -database telegraf -execute "ALTER RETENTION POLICY autogen ON
     telegraf DURATION 168h DEFAULT"

# docker exec -it influxdb influx -database telegraf -execute "show retention policies"
name duration shardGroupDuration replicaN default
---- -------- ------------------ -------- -------
autogen 168h0m0s 168h0m0s 1 true
```

1.7 Useful links

- Running third-party applications on Junos Evolved: https://www.juniper.net/documentation/us/en/software/junos/overview-evo/topics/task/third-party-applications-deploying.html
- Telegraf:https://www.influxdata.com/time-series-platform/telegraf/
- InfluxDB: https://www.influxdata.com/products/influxdb-overview/
- Grafana: https://grafana.com/

1.8 Glossary

- RPD: Routing Process Daemon
- JSD: Juniper Extension Toolkit (JET) Service Process
- TIG: Telegraf, InfluxDB, Grafana
- GNMI: gRPC (Remore Procedure Call) Network Management Interface
- DB: database

- CPU: central processing unit



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